



DSC680

Applied Data Science

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Predicting Digital Experiment Success Using A/B Testing Metrics

Exploring how experimentation and engagement metrics relate to A/B test outcomes through predictive analytics.

Business Problem and Project Objectives



Business Problem

- Organizations increasingly rely on A/B testing to optimize digital products, marketing campaigns, and user experiences
- Despite widespread adoption, experiment outcomes remain difficult to predict consistently
- High failure rates in experimentation lead to wasted resources and missed opportunities for growth
- Predictive analytics may help organizations better understand patterns associated with successful experimentation outcomes.



Project Objectives

- Analyze A/B experimentation metrics and engagement indicators across digital experiments
- Identify key patterns and feature correlations associated with statistically successful test outcomes
- Evaluate and compare machine learning models for predictive performance in forecasting experiment success
- Develop actionable insights to guide resource allocation and experimentation strategy

Dataset Overview



Dataset Summary

1,500

Simulated Digital Experiments

- Synthetically generated to model realistic A/B testing scenarios across digital platforms
- Each record represents a single experiment with measurable performance metrics
- Includes both successful and unsuccessful experiment outcomes as the target variable
- Used to evaluate relationships between experimentation metrics and experiment outcomes



Example Dataset Variables

- **Traffic Volume** — Total visitors exposed to each experiment variant
- **Bounce Rate** — Percentage of single-page sessions without interaction
- **Click-Through Rate** — Ratio of users clicking through to the target action
- **Session Duration** — Average time users spent per session measured in seconds
- **Conversion Rates** — Control and treatment group conversion percentages
- **Statistical Significance** — Whether the observed difference reached the $p < 0.05$ threshold
- **Experiment Outcome** — Binary target label indicating success or failure

Exploratory Data Analysis



Analysis Approach

- Performed systematic exploration of trends, distributions, and relationships across all experimentation metrics
- Assessed feature distributions to detect skewness, outliers, and class imbalance in the target variable
- Examined pairwise relationships between metrics to uncover multicollinearity and interaction effects
- Used descriptive statistics and visual summaries to guide feature selection and modeling decisions
- Identified early patterns suggesting that lift percentage and conversion-related metrics may influence experiment success

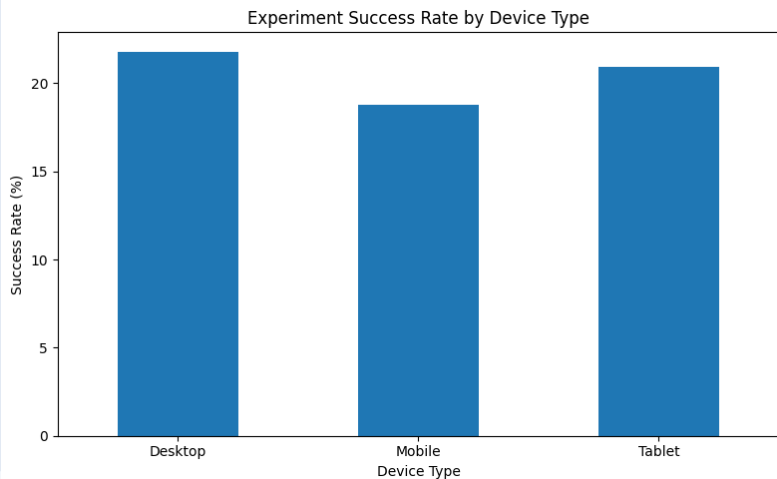


Key Visualizations

- **Device Performance** — Compared experiment success rates across device categories to detect platform-specific patterns
- **Lift Percentage Distributions** — Visualized the spread of treatment lift to understand effect size variation
- **Traffic Relationships** — Explored how traffic volume and session duration interact with conversion outcomes
- **Metric Correlations** — Generated a correlation heatmap to identify strong predictors and redundant features
- **Outcome Distribution** — Assessed class balance between successful and failed experiments to inform modeling strategy

Device Performance and Traffic Relationships

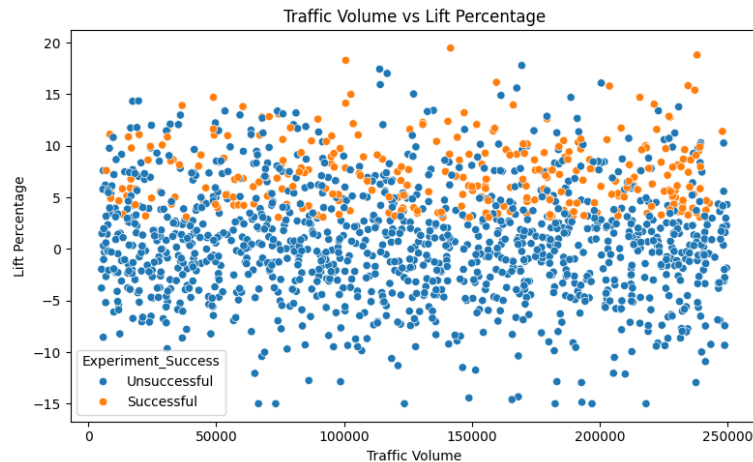
Device Performance



Key Takeaways

- Desktop experiments showed the highest overall success rate
- Mobile experiments demonstrated slightly lower success performance
- Device segmentation may influence experimentation strategy and optimization planning

Traffic Relationships

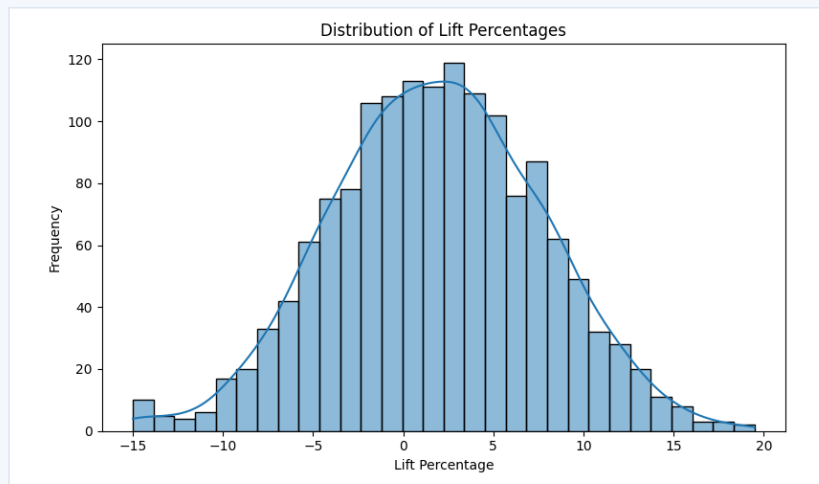


Key Takeaways

- Successful and unsuccessful experiments appeared across all traffic levels
- Higher traffic volume alone did not guarantee stronger experiment lift
- Traffic metrics alone may provide limited predictive value without additional engagement context

Distribution and Correlation Analysis

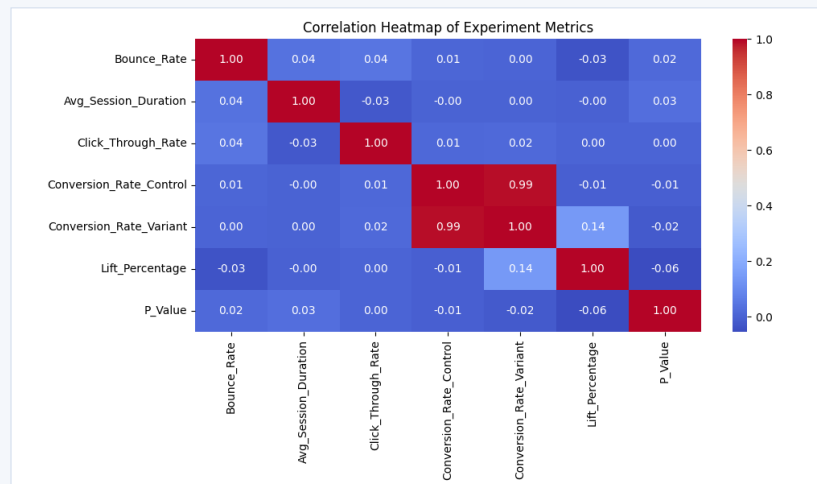
Lift Percentage Distribution



Key Takeaways

- Lift percentages followed a roughly normal distribution centered near modest positive gains
- Both positive and negative experiment outcomes were observed across the dataset
- Most experiments produced relatively small lift changes rather than extreme performance shifts

Metric Correlations



Key Takeaways

- Conversion-related metrics demonstrated the strongest observed correlations
- Most engagement metrics showed relatively weak linear relationships with experiment outcomes
- Correlation analysis helped identify potential redundancy and modeling considerations

Machine Learning Model Evaluation



Logistic Regression

Model Summary

- Baseline classification model used to predict experiment success outcomes
- Provided interpretable results for evaluating metric relationships

Performance Metrics

- **Accuracy:** 78.7%
- **Precision:** 55%
- **Recall:** 9%
- **F1-Score:** 16%

Key Findings

- Correctly identified most unsuccessful experiments
- Struggled to consistently detect successful experiment outcomes
- Demonstrated the impact of class imbalance on predictive performance



Random Forest

Model Summary

- Ensemble classification model used to evaluate non-linear relationships across experimentation metrics
- Applied feature importance analysis to identify influential predictors

Performance Metrics

- **Accuracy:** 78.3%
- **Precision:** 50%
- **Recall:** 2%
- **F1-Score:** 3%

Key Findings

- Produced similar overall accuracy to Logistic Regression
- Demonstrated extremely low recall for successful experiments
- Reinforced the difficulty of predicting experiment success from engagement metrics alone

Target Leakage Discovery and Model Correction



The Problem: Target Leakage

What Happened

- Initial models achieved near-perfect accuracy, signaling potential data integrity issues
- Investigation revealed that certain features directly encoded the target outcome

Leaked Variables Identified

- Conversion Rate Variant
- Conversion Rate Control
- Statistically Significant
- These variables contained information directly tied to the final experiment outcome

Inflated Initial Results

- Initial models achieved unrealistic accuracy scores above 90%
- Random Forest initially produced near-perfect classification performance



The Solution: Corrected Models

Correction Approach

- Removed leakage-related variables from the feature set and retrained both models
- Validated with proper train-test splits using only pre-experiment features

Corrected Performance

- Logistic Regression accuracy decreased to 78.7%
- Random Forest accuracy decreased to 78.3%
- Recall for successful experiments remained very limited

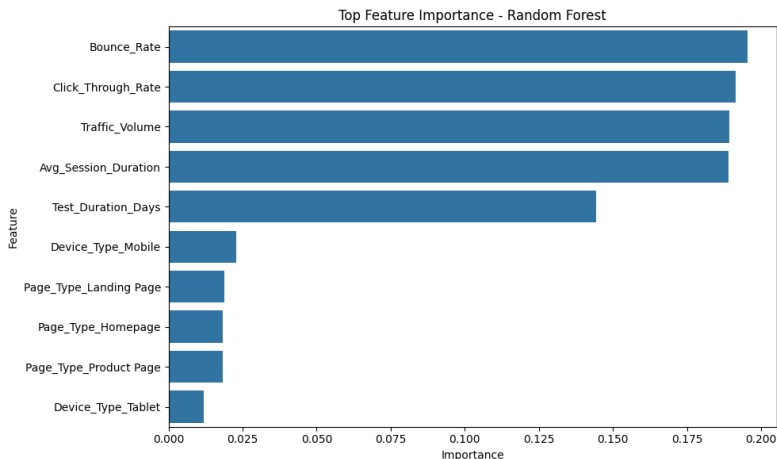
Validation Best Practices

- Always audit features for information that would not be available at prediction time
- Suspiciously high accuracy should trigger immediate leakage investigation
- Realistic validation ensures models generalize to production environments

Feature Importance and Key Findings



Feature Importance



Key Findings

Most Influential Features

- Click-through rate and engagement metrics demonstrated the strongest remaining influence on model behavior
- Session duration and bounce rate contributed moderate predictive value
- Traffic volume alone was not sufficient to reliably predict experiment success

Model Performance Insights

- Random Forest and Logistic Regression produced similar corrected performance results
- Feature importance rankings aligned with expected experimentation behavior patterns

Practical Implications

- Predictive models can help prioritize high-potential experiments, reducing wasted testing resources

Recommendations and Ethical Considerations



Practical Recommendations

- Prioritize engagement and conversion-related metrics when evaluating A/B test performance
- Implement target leakage checks as a standard step in all predictive modeling pipelines to ensure realistic performance evaluation
- Compare multiple modeling approaches to evaluate different behavioral relationships within experimentation data
- Develop experiment prioritization frameworks to help allocate testing resources toward high-potential experiments
- Validate models on real-world production data before deployment to confirm generalizability beyond simulated datasets



Ethical Considerations

Synthetic Data Limitations

- Simulated data may not capture the full complexity of real-world experimentation environments
- Findings should be treated as directional insights requiring validation on production datasets

Modeling Responsibility

- Predictive models should augment, not replace, human judgment in experiment design and decision-making
- Over-reliance on automated predictions risks deprioritizing experiments with long-term strategic value

Responsible Experimentation

- Transparent reporting of model limitations and data provenance is essential for ethical data science practice
- A/B testing must respect user privacy and informed consent, particularly in sensitive digital contexts

Project Conclusion



Summary of Research Outcomes

Project Objective

- This project explored whether machine learning models can reliably predict the success of digital A/B testing experiments using experimentation metrics

Key Analytical Findings

- Corrected models produced moderate overall accuracy scores but struggled to reliably identify successful experiments
- Behavioral metrics such as click-through rate, bounce rate, and session duration demonstrated the strongest remaining influence on model behavior
- Exploratory analysis revealed that experiment outcomes were influenced by multiple interacting factors rather than a single dominant metric

Lessons from Target Leakage

- Initial models achieved unrealistically high accuracy due to target leakage from outcome-derived variables
- Identifying and removing leakage features was a critical step in producing trustworthy, generalizable results

Modeling Limitations

- Corrected models demonstrated limited predictive capability, particularly when identifying successful experiments
- Synthetic data may not fully represent the complexity and variability found in real-world experimentation platforms
- Limited feature set constrained the models; richer behavioral and contextual variables could improve performance

Future Opportunities

- Extend the analysis to real-world A/B testing datasets from production experimentation platforms
- Explore advanced modeling approaches such as gradient boosting and time-series experimentation analysis
- Incorporate richer feature engineering including user segmentation, experiment duration, and seasonal patterns
- Develop an operational scoring tool that helps teams prioritize and allocate experimentation resources efficiently



Questions and Discussion

Thank you for your time and attention.

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